PROBLEM 07

Relative Kinematics & Bar Constraint

Skater Pulled by Vehicle on a Circumference

Original statement · Problems.pdf, p. 8

A skater B is being towed by a vehicle A on a perfectly smooth horizontal plane, where we define an inertial reference frame Oxy. The vehicle is moving along a circumference of radius a and center at O with constant angular velocity ω. The skater is attached to the vehicle by a bar without mass and constant length $l = a$. Initially, the vehicle is located at $(a, 0)$ and the skater is at rest at $(2a, 0)$. Defining a rotating reference frame Ox_1y_1 with origin at O and axis Ox_1 along OA, and calling ψ the angle between the axis Ox_1 and the bar AB, determine:

(a) The relative acceleration of the skater with respect to Ox_1y_1 as a function of ψ and its time derivatives.

(b) The centripetal and Coriolis acceleration terms also as a function of ψ and its time derivatives.

(c) Reduce the evolution of ψ to integrals of the form $\int [...] d\psi = \int [...] dt$ (consider that $\dot{\psi}^2 = \int 2\dot{\psi}\ddot{\psi}\, dt$).

(d) Determine the tension on the bar.

(e) If the rigid bar is substituted by a string which cannot bear compression, determine if/when the string will become loose.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A vehicle A moves counterclockwise at constant speed ω along a circle of radius a. Skater B on a frictionless horizontal plane is towed by vehicle A via a massless bar of length $l = a$.
- **Initial Conditions:** Vehicle A is at $(a, 0)$, skater B is at rest at $(2a, 0)$, meaning at $t = 0$, $\psi(0) = 0$. Because the skater is at rest in the inertial frame, $\dot{\psi}(0) = -2\omega$ (derived in part (c)).
- **Parameters & Units:** Radius and bar length a $[\mathrm{m}]$, angular speed ω $[\mathrm{rad/s}]$, skater mass m $[\mathrm{kg}]$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$.
- **Rotating Frame Attached to OA:** $\mathcal{S}_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, with $\mathbf{i}_1$ pointing from O to vehicle A.

$$\boldsymbol{\omega}_{1/0} = \omega \mathbf{k}_0 = \omega \mathbf{k}_1$$

$$\mathbf{r}_A = a \mathbf{i}_1$$

- **Bar Orientation:** The bar AB is directed at angle ψ relative to $\mathbf{i}_1$:

$$\mathbf{r}_{B/A} = a \cos\psi \, \mathbf{i}_1 + a \sin\psi \, \mathbf{j}_1$$

- **Position of Skater B in Rotating Frame:**

$$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A} = a(1+\cos\psi)\mathbf{i}_1 + a\sin\psi\,\mathbf{j}_1$$

- **Change of basis $\mathcal{B}_0 \to \mathcal{B}_1$.** The vehicle starts at $(a,0)$ and turns counterclockwise at rate ω, so $\mathbf{i}_1 = \cos\omega t\,\mathbf{i}_0 + \sin\omega t\,\mathbf{j}_0$ and the columns of the matrix below are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- **Bar basis $\mathcal{B}_1 \to \mathcal{B}_{\text{bar}}$.** The pair $\mathbf{u}_{AB} = \cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1$, $\mathbf{u}_{\perp} = -\sin\psi\,\mathbf{i}_1 + \cos\psi\,\mathbf{j}_1$ used in parts (c) and (d) is obtained with

$$[{}_1R_{\text{bar}}] = \begin{pmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_1R_{\text{bar}}] = 1, \qquad [{}_1R_{\text{bar}}][{}_1R_{\text{bar}}]^T = I$$

- The composition $[{}_0R_{\text{bar}}] = [{}_0R_1][{}_1R_{\text{bar}}]$ is the planar rotation by $\omega t + \psi$ (the bar direction measured from the inertial x axis). The determinants and the products RR^T were verified symbolically with sympy.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Relative Acceleration in $\mathcal{S}_1$

Pedagogical Justification: The relative velocity and acceleration are the time derivatives of $\mathbf{r}_B$ observed within frame $\mathcal{S}_1$, treating unit vectors $\mathbf{i}_1, \mathbf{j}_1$ as constant.

- **Relative Velocity:**

$$\mathbf{v}_{\text{rel}} = \frac{d\mathbf{r}_B}{dt}\Big|_{\mathcal{S}_1} = -a\dot{\psi}\sin\psi\,\mathbf{i}_1 + a\dot{\psi}\cos\psi\,\mathbf{j}_1$$

- **Relative Acceleration (Explicit Product Rule):**

$$\mathbf{a}_{\text{rel}} = \frac{d\mathbf{v}_{\text{rel}}}{dt}\Big|_{\mathcal{S}_1} = \frac{d}{dt}(-a\dot{\psi}\sin\psi)\mathbf{i}_1 + \frac{d}{dt}(a\dot{\psi}\cos\psi)\mathbf{j}_1$$

$$= -a(\ddot{\psi}\sin\psi + \dot{\psi}^2\cos\psi)\mathbf{i}_1 + a(\ddot{\psi}\cos\psi - \dot{\psi}^2\sin\psi)\mathbf{j}_1$$

(b) Coriolis and Centripetal Accelerations

> **Pedagogical Justification:** By Coriolis' theorem of relative kinematics (Notes Sec. 3.2, Eq. 3.27):

$$\mathbf{a}_{\text{cor}} = 2\boldsymbol{\omega}_{1/0} \times \mathbf{v}_{\text{rel}}$$

$$\mathbf{a}_{\text{cf}} = \boldsymbol{\omega}_{1/0} \times (\boldsymbol{\omega}_{1/0} \times \mathbf{r}_B) = -\omega^2 \mathbf{r}_{B\perp}$$

- **Coriolis Calculation:**

$$\mathbf{a}_{\text{cor}} = 2(\omega\mathbf{k}_1) \times (-a\dot{\psi}\sin\psi\,\mathbf{i}_1 + a\dot{\psi}\cos\psi\,\mathbf{j}_1)$$

Using $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$ and $\mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$:

$$\mathbf{a}_{\text{cor}} = 2\omega \left(-a\dot{\psi} \sin\psi \, \mathbf{j}_1 - a\dot{\psi} \cos\psi \, \mathbf{i}_1 \right) = -2a\omega\dot{\psi}(\cos\psi \, \mathbf{i}_1 + \sin\psi \, \mathbf{j}_1)$$

(Compare with the official solution key (Problems.pdf, Solution keys section).)

- **Centripetal / Centrifugal Acceleration Term:**

Since $\mathbf{r}_B$ lies entirely in the x_1y_1 plane, $\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_B) = -\omega^2 \mathbf{r}_B$:

$$\mathbf{a}_{\text{cf}} = -\omega^2 \mathbf{r}_B = -a\omega^2[(1 + \cos\psi)\mathbf{i}_1 + \sin\psi \, \mathbf{j}_1]$$

(Compare with the official solution key (Problems.pdf, Solution keys section).)

(c) Reduction of Equation of Motion to Integrals (Quadrature)

> **Pedagogical Justification:** The only force acting on skater B in the horizontal plane is the tension $\mathbf{T}$ from the bar, which acts along the bar: $\mathbf{u}_{AB} = \cos\psi \, \mathbf{i}_1 + \sin\psi \, \mathbf{j}_1$. Therefore, the total acceleration of B in the direction perpendicular to the bar $\mathbf{u}_\perp = -\sin\psi \, \mathbf{i}_1 + \cos\psi \, \mathbf{j}_1$ must vanish: $\mathbf{a}_B \cdot \mathbf{u}_\perp = 0$.

- **Projection of Acceleration:**

$$\mathbf{a}_{\text{total}} = \mathbf{a}_{\text{rel}} + \mathbf{a}_{\text{cor}} + \mathbf{a}_{\text{cf}}$$

Taking the dot product with $\mathbf{u}_\perp$:

- $\mathbf{a}_{\text{rel}} \cdot \mathbf{u}_\perp = a\ddot{\psi}(\sin^2\psi + \cos^2\psi) = a\ddot{\psi}$.
- $\mathbf{a}_{\text{cor}} \cdot \mathbf{u}_\perp = -2a\omega\dot{\psi}(\cos\psi(-\sin\psi) + \sin\psi\cos\psi) = 0$.
- $\mathbf{a}_{\text{cf}} \cdot \mathbf{u}_\perp = -a\omega^2[(1 + \cos\psi)(-\sin\psi) + \sin\psi\cos\psi] = -a\omega^2(-\sin\psi - \sin\psi\cos\psi + \sin\psi\cos\psi) = a\omega^2\sin\psi$.

Therefore:

$$a\ddot{\psi} + a\omega^2 \sin\psi = 0 \implies \ddot{\psi} + \omega^2 \sin\psi = 0$$

Multiplying by $2\dot{\psi}$ and using the chain rule $\frac{d}{dt}(\dot{\psi}^2) = 2\dot{\psi}\ddot{\psi}$ and $\frac{d}{dt}(\cos\psi) = -\sin\psi \, \dot{\psi}$:

$$2\dot\psi\ddot\psi = -2\omega^2 \sin\psi\,\dot\psi \implies \frac{d}{dt}(\dot\psi^2) = -2\omega^2\sin\psi\,\dot\psi = 2\omega^2\frac{d}{dt}(\cos\psi)$$

Initial conditions. The skater starts at rest in the inertial frame, $\mathbf{v}_B(0) = \mathbf{0}$, at $\mathbf{r}_B(0) = 2a\,\mathbf{i}_1$ (so $\psi(0) = 0$). Since $\mathbf{v}_B = \boldsymbol{\omega}_{1/0}\times\mathbf{r}_B + \mathbf{v}_{\text{rel}}$:

$$\mathbf{v}_{\text{rel}}(0) = -\omega\mathbf{k}_1 \times 2a\,\mathbf{i}_1 = -2a\omega\,\mathbf{j}_1, \qquad \mathbf{v}_{\text{rel}}(0) = a\dot\psi(0)\,\mathbf{j}_1 \implies \dot\psi(0) = -2\omega$$

Integrating with Barrow's rule between $t = 0$ ($\psi = 0$, $\dot\psi = -2\omega$) and a generic instant, with u a dummy variable ($du = \dot\psi\,dt$ along the motion):

$$\dot\psi^2 - (-2\omega)^2 = 2\omega^2\int_0^{\psi}(-\sin u)\,du = 2\omega^2\,[\cos u]_0^{\psi} = 2\omega^2(\cos\psi - 1)$$

$$\dot\psi^2 = 2\omega^2(1+\cos\psi) = 4\omega^2\cos^2\!\left(\frac{\psi}{2}\right) \quad \text{using } 1+\cos\psi = 2\cos^2(\psi/2)$$

Taking the square root, $\dot\psi = \pm 2\omega\cos(\psi/2)$. For $-\pi < \psi < \pi$ the factor $\cos(\psi/2)$ is positive, so the sign cannot change, and $\dot\psi(0) = -2\omega$ selects the minus sign (ψ decreases from zero):

$$\dot\psi = -2\omega\cos\left(\frac{\psi}{2}\right) \quad \text{for } -\pi < \psi \le 0$$

This agrees with the official key (Problems.pdf, PDF page 72).

Separating variables and integrating from $(t,\psi) = (0,0)$, with the substitution $w = u/2$, $du = 2\,dw$:

$$\int_0^{\psi} \frac{du}{2\cos(u/2)} = -\omega \int_0^{t} dt' \implies \int_0^{\psi/2} \frac{dw}{\cos w} = \ln\left|\sec\frac{\psi}{2} + \tan\frac{\psi}{2}\right| = -\omega t$$

(d) Tension on the Bar

> **Pedagogical Justification:** The tension is obtained from the radial force balance along the bar:

$$T = -m(\mathbf{a}_{\text{total}} \cdot \mathbf{u}_{AB})$$

Evaluating the terms along $\mathbf{u}_{AB} = \cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1$:

- $\mathbf{a}_{\text{rel}} \cdot \mathbf{u}_{AB} = -a(\ddot{\psi}\sin\psi + \dot{\psi}^2\cos\psi)\cos\psi + a(\ddot{\psi}\cos\psi - \dot{\psi}^2\sin\psi)\sin\psi = -a\dot{\psi}^2$.
- $\mathbf{a}_{\text{cor}} \cdot \mathbf{u}_{AB} = -2a\omega\dot{\psi}(\cos^2\psi + \sin^2\psi) = -2a\omega\dot{\psi}$.
- $\mathbf{a}_{\text{cf}} \cdot \mathbf{u}_{AB} = -a\omega^2[(1+\cos\psi)\cos\psi + \sin^2\psi] = -a\omega^2(1+\cos\psi)$.

$$T = ma\left[\dot{\psi}^2 + 2\omega\dot{\psi} + \omega^2(1+\cos\psi)\right]$$

With $c = \cos(\psi/2)$: $\dot{\psi}^2 = 4\omega^2 c^2$, $\dot{\psi} = -2\omega c$, $1+\cos\psi = 2c^2$, hence $T = ma\omega^2(4c^2 - 4c + 2c^2) = ma\omega^2(6c^2 - 4c)$:

$$T = 2ma\omega^2\cos\left(\frac{\psi}{2}\right)\left(3\cos\left(\frac{\psi}{2}\right) - 2\right)$$

(Compare with the official solution key (Problems.pdf, Solution keys section).)

(e) Loose String Condition

Pedagogical Justification: A flexible string cannot support compression ($T \geq 0$). The string becomes loose when the tension drops to zero: $T = 0$.

$$3\cos\left(\frac{\psi}{2}\right) - 2 = 0 \implies \cos\left(\frac{\psi}{2}\right) = \frac{2}{3}$$

$$\frac{\psi}{2} = -\arccos\left(\frac{2}{3}\right) \implies \psi = -2\arccos\left(\frac{2}{3}\right) \approx -96.38^\circ$$

(Compare with the official solution key (Problems.pdf, Solution keys section).)

PHASE 4

Physical Interpretation, Limiting Cases and Dimensional Check

- **Dimensional check:**
- $[\ddot{\psi}] = [\omega^2 \sin\psi] = \mathrm{s}^{-2}$ and $[\dot{\psi}^2] = [2\omega^2(1+\cos\psi)] = \mathrm{s}^{-2}$ (the angle is dimensionless).
- $[T] = [m\,a\,\omega^2] = \mathrm{kg}\cdot\mathrm{m}\cdot\mathrm{s}^{-2} = \mathrm{N}$.
- **Limit $\omega \to 0$:** the vehicle is at rest, the skater is at rest, and $T \to 0$, $\dot{\psi} \to 0$, as expected.
- **Independent check of T at $t = 0$ ($\psi = 0$).** The closed form gives $T(0) = 2ma\omega^2 \cdot 1 \cdot (3-2) = 2ma\omega^2$. For a rigid bar of length $l = a$, the relative velocity $\mathbf{v}_B - \mathbf{v}_A$ is perpendicular to the bar, hence $(\mathbf{a}_B - \mathbf{a}_A)\cdot\mathbf{u}_{AB} = -\|\mathbf{v}_B - \mathbf{v}_A\|^2/a$. At $t = 0$: $\mathbf{v}_B = \mathbf{0}$, $\|\mathbf{v}_A\| = a\omega$, $\mathbf{a}_A = -a\omega^2\mathbf{i}_0$ and $\mathbf{u}_{AB} = \mathbf{i}_0$, so $\mathbf{a}_B\cdot\mathbf{u}_{AB} = -a\omega^2 - a\omega^2 = -2a\omega^2$ and $T = 2ma\omega^2$ (Consistent).
- **Symbolic check (sympy).** Differentiating twice, in the inertial frame, the position $\mathbf{r}_B = [{}_0R_1]\,a\,(1+\cos\psi,\ \sin\psi,\ 0)^T$ and projecting on $\mathbf{u}_{AB}$ and $\mathbf{u}_\perp$ gives $\mathbf{a}_B\cdot\mathbf{u}_\perp = a(\ddot{\psi} + \omega^2\sin\psi)$ and $T = ma\left[\dot{\psi}^2 + 2\omega\dot{\psi} + \omega^2(1+\cos\psi)\right]$, identical to parts (c) and (d); substituting $\dot{\psi} = -2\omega\cos(\psi/2)$ reproduces $T = 2ma\omega^2\cos(\psi/2)\,(3\cos(\psi/2) - 2)$ with zero residual.
- **When the string goes loose (numerical).** The closed form $\omega t = -\ln|\sec(\psi/2) + \tan(\psi/2)|$ evaluated at $\cos(\psi/2) = 2/3$ gives $\omega t_{\text{loose}} = \ln\dfrac{2}{3-\sqrt{5}} = \ln\dfrac{3+\sqrt{5}}{2} = 0.96242$. An RK4 integration (step 10^{-5}, $\omega = 1$) of $\ddot{\psi} = -\omega^2\sin\psi$ with

$\psi(0)=0, \dot\psi(0)=-2\omega$ reaches $\cos(\psi/2)=2/3$ at $\omega t = 0.96242$ with $\psi=-96.379^\circ$, in agreement with $-2\arccos(2/3)=-96.379^\circ$; there $\dot\psi^2 = 1.777787\,\omega^2$ against $2\omega^2(1+\cos\psi)=1.777787\,\omega^2$.

Source: Problems.pdf (Page 8)